Machine basics, DC machines, rotating fields

Answers and worked solutions · Day 07 · Fri 30 Oct · pack M6

5. Practice

From your course files

Jun24 Q3a (past exam, 6 marks).

The official answer is only “synchronous, DC, induction; sketch curves from Sections 4.1 and 7.9; halve V and f / halve V / halve V and f”. (On the paper the three plots are stacked top to bottom; above they are shown left to right in the same order.)

(i) [3 marks]

1. **Synchronous machine:** a vertical line at n_S. At half speed it is a vertical line at $n_S/2$.
2. **DC machine:** a line sloping down from n_{NL} at τ = 0. At half speed the new line starts at $n_{NL}/2$ and is **parallel** to the original. The speed drop at any torque is unchanged, so the curve slides left.
3. **Induction machine:** a steep line sloping down from n_S, shown in its linear region near n_S. At half speed it starts at $n_S/2$ and is **parallel** to the original. The slip speed at a given torque is unchanged.

The figure in Section 1.3 shows all three, with the halved versions dashed.

(ii) [3 marks]

- **Synchronous:** halve the supply frequency f, which halves $n_S = 60f/p$. **Also halve V**, to keep V/f and hence the peak flux constant. This avoids core saturation and large currents.
- **DC: halve the terminal voltage V.** The no-load speed V/k halves.
- **Induction: halve both f and V** (constant V/f), for the same reason as the synchronous machine.

In practice the AC machines need a variable-voltage, variable-frequency inverter. The DC machine needs a variable DC supply, such as a PWM chopper like the cordless-drill example in Lecture 4.1.6.

PQ 4.3 (practice questions).

a. The rating is the output: **500 W**.

b. $\omega_R = 10\,000 \times 2\pi/60 = 1047.2$ rad/s, so $\tau_R = 500/1047.2 =$ **0.4775 Nm**.

c. The speed drops 2000 rpm at rated torque, so only 1000 rpm at half torque: $n = 12\,000 - 1000 =$ **11 000 rpm**.

d. $P = \tau\omega = (0.4775/2)(11\,000 \times 2\pi/60) =$ **275 W**. The torque is half, but the speed is higher than rated speed, so the power is more than half of rated power.

Course answers: 500 W, 0.4775 Nm, 11 000 rpm, 275 W. All agree.

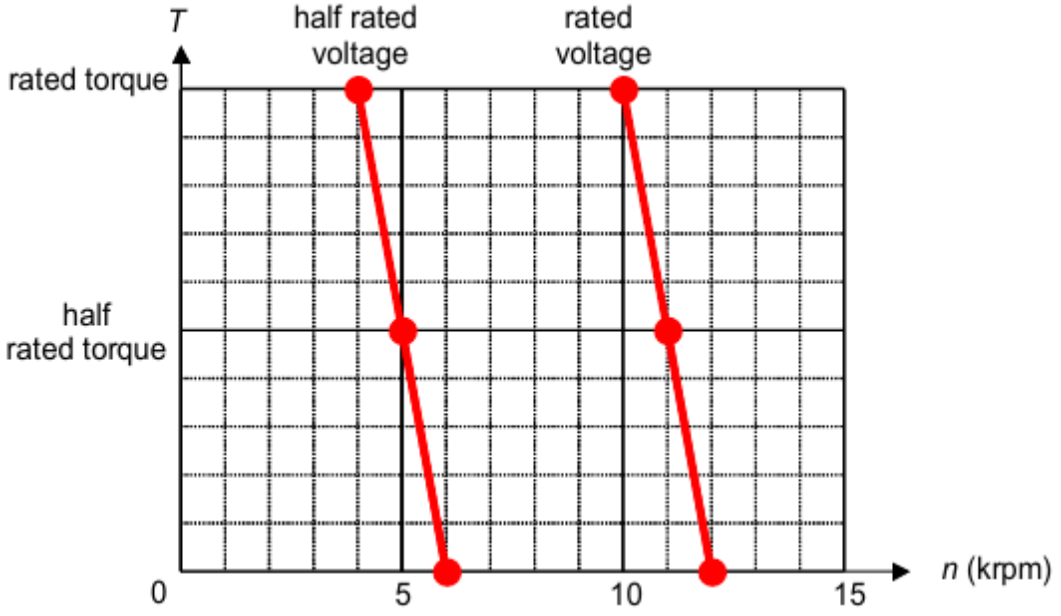
PQ 4.4 (practice questions).

a. The no-load speed is proportional to the supply voltage: $12\,000/2 =$ **6000 rpm**.

b. The 2000 rpm drop at full-load torque does not depend on the voltage: $6000 - 2000 =$ **4000 rpm**.

c. Half torque gives half the drop: $6000 - 1000 =$ **5000 rpm**.

d. Two parallel straight lines: rated voltage from (12 krpm, 0) through (11 krpm, half) to (10 krpm, rated); half voltage from (6 krpm, 0) through (5 krpm, half) to (4 krpm, rated).



PQ 4.4 official answer sketch (from the practice questions): the rated- and half-rated-voltage lines with the three operating points on each

PQ 4.7 (practice questions).

a. $n_S \propto f$, so $f = 60 \times 900/1200 =$ **45 Hz**. Keep V/f constant: $V = 110 \times 45/60 =$ **82.5 V**.

b. The slip speed at a given torque stays the same: 100 rpm at rated torque, 50 rpm at half torque.

c. $900 - 100 =$ **800 rpm**. (ii) $900 - 50 =$ **850 rpm**.

Course answers: 82.5 V, 45 Hz; 800 rpm, 850 rpm.

PQ 4.11 (practice questions).

> $p = 60f/n_S = 60 \times 50/630 = 4.762$ pole-pairs. The closest integer is 5 pole-pairs, so **10 poles**, giving $n_S = 60 \times 50/5 =$ **600 rpm**. (8 poles gives 750 rpm and 12 poles gives 500 rpm, both further away.) Here you round to the nearest, because the question asks for the closest speed, not a speed above a rated speed.

PQ 4.13 (practice questions).

> a. $n_S = 120 \times 50/8 =$ **750 rpm**.
> b. A synchronous machine runs at constant speed: **750 rpm**.
> c. $\tau_R = 11\,000/(750 \times 2\pi/60) = 11\,000/78.54 =$ **140.1 Nm**.
> d. Still **750 rpm**.
> e. The speed is constant, so the power is proportional to the torque: **5.5 kW**.

PQ 5.8 (practice questions).

> $k = 0.24$ V/(rad/s), $R = 0.5\ \Omega$, $V = 12$ V.
>
> a. No load: τ = 0, so I = 0 and E = V. $\omega_{NL} = 12/0.24 = 50$ rad/s, i.e. **477.5 rpm**.
>
> b. $\eta = n/n_{NL} = 400/477.5 =$ **83.77 %**.
>
> c. At stall ω = 0: (i) $E =$ **0 V**; (ii) $I = V/R =$ **24 A**; (iii) $\tau = kI =$ **5.760 Nm**; (iv) $P_{IN} = VI =$ **288 W**; (v) $P_{OUT} =$ **0 W** (n = 0); (vi) **0 %**.
>
> d. $I = 0.2/0.24 = 0.8333$ A, $E = 12 - 0.8333 \times 0.5 = 11.58$ V, $\omega = 11.58/0.24 = 48.26$ rad/s, so **460.9 rpm**.
>
> For reference, PQ 5.7: $E = 0.24 \times 41.89 = 10.05$ V, $I = (12 - 10.05)/0.5 = 3.894$ A (official 3.9 A, rounded), $\tau = 0.936$ Nm, $P_{OUT} \approx 39.2$ W, $P_{IN} = 46.8$ W, η = 83.78 % / 83.75 % (the official answers differ only by that rounding of I).

PQ 5.10 (practice questions).

> a. $k = \tau/I = 15/35 = 0.4286$ Nm/A (= V/(rad/s)). At no load $E = V$, so $\omega_{NL} = 24/0.4286 = 56$ rad/s, i.e. **534.8 rpm**.
>
> b. At stall E = 0, so $R = V/I = 24/35 =$ **0.6857 Ω**.
>
> c. At 400 rpm: $E = 0.4286 \times (2\pi \times 400/60) = 17.95$ V, $I = (24 - 17.95)/0.6857 = 8.820$ A, $P = EI =$ **158.3 W**.

> **NOTE:** the PQ answer (158.4 W) and the WS5 handwritten answer (158.1 W) both differ only through intermediate rounding. Any of these will get the marks.

PQ 5.19 (practice questions).

a. $k = E/\omega = 10.5/(150 \times 2\pi/60) = 10.5/15.71 =$ **0.6684 V/(rad/s)**.

b. $E = 14.6$ V: $\omega = 14.6/0.6684 = 21.84$ rad/s = **208.6 rpm**. With $E = V$ the current is **0 A**.

c. Generator: $E = V + IR = 14.6 + 10 \times 0.4 = 18.6$ V, so $\omega = 18.6/0.6684 = 27.83$ rad/s = **265.7 rpm**.

d. For a DC generator $\eta = V/E = 14.6/18.6 =$ **78.49 %**.

e. At low wind speed $E < 14.6$ V, so without the diode current would flow from the battery into the machine. That discharges the battery and makes the machine run as a motor, turning the wind turbine into a fan. The diode blocks this reverse current.

PQ 6.4 (practice questions).

Read the order in which the phases reach their positive peaks.

a. V_A, then V_C, then V_B: **acb** (or cba or bac).
b. V_A, then V_B, then V_C: **abc** (or bca or cab).

PQ 6.5 (practice questions).

Swap any two of the three supply leads. This reverses the phase sequence, so the rotating field (and the rotor) turns the other way. Relabelling the phases cyclically does not change the sequence.

PQ 6.7 (practice questions).

Poles come in N–S pairs, so the number must be an even integer: (a) **yes** (b) **no** (odd) (c) **no** (not an integer) (d) **yes**.

PQ 6.9 (practice questions).

a. The grid is 50 Hz and the fewest possible poles is 2 ($p = 1$): $n_S = 60 \times 50/1 =$ **3000 rpm, two poles**.

b. $p = 60f/n_S = 60 \times 50/100 = 30$ pole-pairs, so **60 poles**.

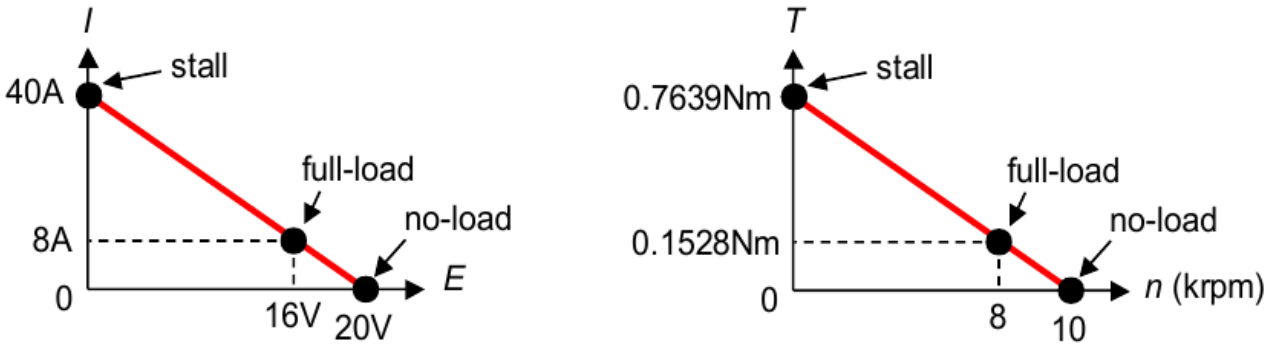
PQ 5.11 (practice questions, extra).

$k = 20/(2\pi \times 10\,000/60) = 0.01910$ V/(rad/s).

a. Zero speed: E = 0, so $I = V/R = 20/0.5 =$ **40 A** and $\tau = kI =$ **0.7639 Nm**.

b. At 8000 rpm $E = 20 \times 8000/10\,000 = 16$ V, so $I = (20 - 16)/0.5 =$ **8 A** and $\tau = kI =$ **0.1528 Nm**.

c. Both characteristics are straight lines: I falls from 40 A at E = 0 (stall) to 0 at E = 20 V (no load), passing 8 A at 16 V (full load); τ falls from 0.7639 Nm at n = 0 to 0 at 10 krpm, passing 0.1528 Nm at 8 krpm.

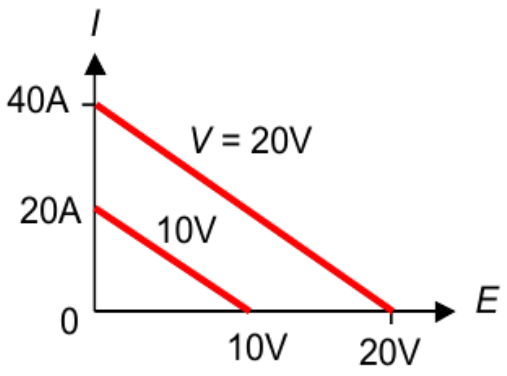


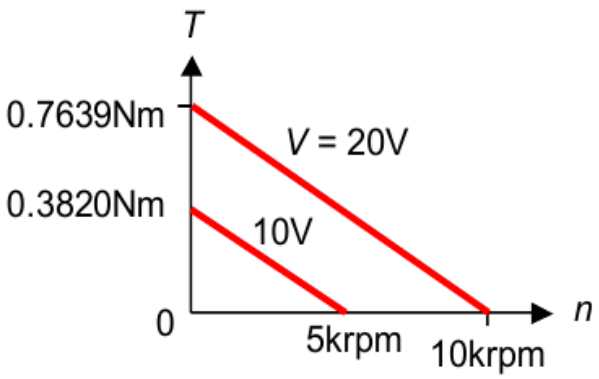
PQ 5.11 official answer (from the practice questions): the I–E and torque–speed lines with stall, full-load and no-load points

d. Half rated torque is half the full-load current and half the speed drop: **4 A, 9000 rpm**.

PQ 5.12 (practice questions, extra).

The rated-voltage lines come from PQ 5.11 (stall 40 A and 0.7639 Nm; no load 20 V and 10 krpm). At half voltage (10 V) the no-load induced voltage and speed halve (10 V, 5 krpm), and the stall current V/R halves to **20 A**, so the stall torque halves to **0.3820 Nm**. The half-voltage lines are parallel to the rated ones.





PQ 5.12 official answer (from the practice questions): rated (20 V) and half (10 V) voltage lines on the I–E and torque–speed axes

New questions

N1. [4 marks]

a. $p = 60 \times 60/1140 = 3.158$, which rounds down to p = 3. So $n_S = 3600/3 = \mathbf{1200\ rpm}$, with **6 poles**.

b. $\eta_{max} = n_R/n_S = 1140/1200 = \mathbf{95.0\ \%}$.

c. The slip speed at rated torque is 60 rpm. $n = 1200 - 0.6 \times 60 = \mathbf{1164\ rpm}$.

N2. [6 marks]

a. $\omega_{NL} = 2\pi(3000)/60 = 314.16$ rad/s. At no-load, E = V, so $k = 48/314.16 = \mathbf{0.1528\ V/(rad/s)}$.

b. $\tau_R = kI_R = 0.1528 \times 12 = \mathbf{1.833\ Nm}$. $E_R = 48 - 12(0.25) = 45$ V, so $\omega_R = 45/0.1528 = 294.5$ rad/s, which is **2812.5 rpm**. $\eta = E/V = 45/48 = \mathbf{93.75\ \%}$. Check: $P_{out} = EI = 540$ W and $P_{in} = 576$ W.

c. $I_S = V/R = 48/0.25 = \mathbf{192\ A}$. $\tau_S = kI_S = \mathbf{29.34\ Nm}$, which is **16 times** τ_R (= 192/12).

d. The maximum is at $n_{NL}/2 = \mathbf{1500\ rpm}$, where $I = I_S/2 = 96$ A and E = 24 V. So $P_{max} = 24 \times 96 = \mathbf{2304\ W}$. That is 8 times the rated current, at only 50 % efficiency, so the motor would overheat.

N3. [6 marks] (chains from N2)

a. $n_{NL} \propto V$: **1500 rpm**. Rated torque means I = 12 A still, so $E = 24 - 12(0.25) = 21$ V, giving $\omega = 21/0.1528 = 137.4$ rad/s, which is **1312.5 rpm**. (Check: the drop is

187.5 rpm, the same as at 48 V.) $\eta = E/V = 21/24 = \mathbf{87.5\,\%}$. The same I^2R loss is now a bigger fraction of a smaller power.

b. $E = k\omega = 0.1528 \times 2\pi(1000)/60 = 16.0$ V. So $V = E + IR = 16 + 3 = \mathbf{19.0\ V}$.

c. I = 6 A, so $E = 24 - 1.5 = 22.5$ V, which gives **1406 rpm**. (Equivalently, $1500 - 0.5 \times 187.5 = 1406.25$ rpm.)

N4. [6 marks]

a. I = 0 when E = V: $\omega = 24/0.5 = 48$ rad/s, which is **458.4 rpm**.

b. Generating, so $E = V + IR = 24 + 5(0.8) = 28$ V. Then $\omega = 56$ rad/s, which is **534.8 rpm**. $\tau = kI = \mathbf{2.5\ Nm}$ (input torque). $P_M = EI = \mathbf{140\ W}$. $P_E = VI = \mathbf{120\ W}$. $\eta = V/E = \mathbf{85.71\,\%}$.

c. $E = 0.5 \times 2\pi(400)/60 = 20.94$ V < 24 V. The current flows **from the battery into the machine**: $I = (24 - 20.94)/0.8 = \mathbf{3.820\ A}$. The torque is $kI = \mathbf{1.910\ Nm}$ of *motoring* torque. The battery is being discharged (91.7 W) to spin the turbine as a fan. A series diode prevents this (Lecture 5.3.2, PQ 5.19e).

N5. [5 marks]

a. The slip speed at rated torque is $1500 - 1440 = 60$ rpm, and it stays constant. So the required $n_S = 900 + 60 = 960$ rpm. $f = 960 \times p/60 = 960 \times 2/60 = \mathbf{32.0\ Hz}$, and $V = 415 \times 32/50 = \mathbf{265.6\ V}$.

b. $n = 960 - 30 = \mathbf{930\ rpm}$.

c. A synchronous machine has no slip, so $n_S = 900$ rpm. $f = 900 \times 3/60 = \mathbf{45\ Hz}$, and $V = 415 \times 45/50 = \mathbf{373.5\ V}$.

N6. [4 marks] (words only)

a. **Still clockwise.** The terminal sequence A, B, C now sees b, c, a, which is a cyclic shift of abc and so the same phase sequence. Only swapping *two* leads (giving acb) reverses the field and the rotor.

b. $p = 60f/n_S = 60 \times 50/100 = 30$ pole-pairs, so **60 poles**. At 60 Hz, p = 36, so **72 poles**.

c. **2.** One electrical cycle moves the field by one pole-pair, which is half a revolution in a 4-pole machine. The field turns at $n_S = 1500$ rpm while the current cycles at 3000 cycles per minute.

6. Quick recall check

Question

1. $n_S = 60 \times 60/4 = 900$ rpm.
2. $n_{NL} = 1500$ rpm. $n_R = 1200$ rpm (the drop of 300 rpm is unchanged).
3. E = 0 and $I_S = V/R$, which is very large. $\tau_S = kI_S$.
4. Peak flux is proportional to V/f (from $E = 4.44Nf\hat{\Phi}$). If f falls with V fixed, the flux rises, the core saturates and the current becomes excessive.
5. p = 2 (60 × 50/1470 = 2.04), so $n_S = 1500$ rpm and $\eta = 1470/1500 = 98\,\%$.